Scalability & Future Plan

Date	1 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines the scalability and future enhancement plans for the Flood Prediction System, including cloud deployment, live weather data integration, improved security, and support for larger user bases and datasets.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses historical datasets only; no real-time weather data integration	Reduces prediction accuracy for live scenarios	High
2	Application is deployed locally and cannot be accessed remotely	Limits availability to end users	High
3	No user authentication or alert notification system	Limits security and practical usability	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited local users	Deploy on cloud platforms (IBM Cloud/AWS/Azure) with scalable infrastructure
2	Data Storage	CSV-based dataset storage	Integrate a cloud database such as MySQL or MongoDB
3	Performance	Local model inference	Optimize the ML model and expose prediction services through REST APIs
4	Security	Basic input validation	Implement user authentication, HTTPS, and role-based access control

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Integration with Live Weather APIs	Within 2 Months	Provides real-time flood prediction
Phase 2	SMS/Email Alert Notification System	Within 3 Months	Enables timely flood warnings to users
Phase 3	Cloud Deployment and Mobile Application	Within 6 Months	Improves accessibility, scalability, and user experience